Multimodal Biometric Authentication Using Keystroke Dynamics and Speech Biometrics as an Alternative to Traditional 2FA

A Design Project

Presented to the Faculty of Electrical Engineering Department

College of Engineering

Batangas State University

Alangilan Campus

In Partial Fulfillment

Of the Requirements for Degree

Bachelor of Science in Computer Engineering

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CHAPTER 1

INTRODUCTION

BACKGROUND OF THE STUDY

Many online systems still continue to use passwords as the main form of authentication, even though they are highly prone to data breaches and unauthorized access. To mitigate risks and attacks on the authentication process, two-factor authentication (2FA) has emerged as a solid protection by adding an extra layer of defense. This authentication can be implemented through different methods, most of which includes the use of one-time passwords sent via SMS, hardware tokens, or OTPs produced by algorithms. However, this traditional 2FA method is still observed to present challenges such as SMS delays, unavailability of service in out-of-coverage areas, code expiration, token drawbacks and inconvenience, and potential SMS costs (Omwoyo et al., 2022). Biometric authentication, as much as it has advantages, cannot stand alone because it may be spoofed from misuse and leakage of biometric data or be affected by environmental factors such as poor lighting, unclean fingers, and the use of unstable low-end devices that challenges accurate scanning (Hussaini, 2024). This has brought the need for an alternative authentication system that enhances security at the same time without compromising ease of use and without relying on obsolete or easily explorable techniques.

Several existing technologies aim to solve these authentication challenges by combining multiple layers of security and enhancing user convenience, finding alternative authentication methods that extend beyond traditional passwords and standard two-factor authentication (2FA)

techniques. One promising way to improve digital security is by combining different types of biometric authentication, like how they type or speak, to confirm their identity. Behavioral biometrics are increasingly favored since they are less expensive to implement and may unobtrusively gather data in the background without disrupting the user. Research like Ryu et al. (2021) shows that using multiple biological methods improves the accuracy of identification and makes users feel less uncomfortable than relying on a single method to make the system more reliable and resistant to scam.

Through the use of keystrokes dynamics and speech biometrics, the study endeavors to develop a multimodal biometric authentication system that enhances the level of security while simultaneously making the procedure of logging in more convenient. Speech biometrics examines aspects of the voice such as pitch and tone. Keystroke dynamics investigates the speed and rhythm of typing. The combination of these two components enhances the resilience and adaptability of the system to reduce the risk of session hijacking and other types of attacks.

This system will make use of machine learning techniques like Random Forest algorithms applied to keystrokes data. In order to extract speech characteristics, Mel-Frequency Cepstral Coefficients and Convolutional Neural Networks will be utilized in order to investigate these characteristics. The fact that this system has an ongoing identification function that validates the user's identity many times during the session is one of the things that sets it apart from other similar systems; thus, Support Vector Machine will be used in order to identify anomalous activity. These algorithms will be able to detect deviations from the regular behaviors of the user and will ask the user to re-authenticate themselves when it is required.

Keystroke and speech biometrics work well together which makes the system more accurate and safe while making things easy for the user. Since both biometrics are based on normal user behavior, the system works in the background without bothering the user, and multimodal systems like this are also harder for hackers to fool and are more stable even in noisy or changing environments. This will make a good fit for online security, where conditions can change quickly.

RESEARCH QUESTION

The main objective of this study is to develop a multimodal biometric authentication system that integrates keystroke dynamics and speech biometrics to enhance authentication security.

The research will focus on addressing the following questions:

1. What impact does integrating keystroke dynamics and speech biometrics have on the accuracy and reliability of authentication systems?
2. What are the essential characteristics and procedures for collecting comprehensive datasets of keystroke dynamics and speech biometrics to support authentication?
3. How does combining two behavioral biometrics increase authentication security compared to single-modal systems?
4. What evaluation methods can be used to assess the security performance of the proposed multimodal biometric authentication system against potential threats and vulnerabilities?

RESEARCH OBJECTIVES

The main objective of this study is to develop a multimodal biometric authentication system that combines keystroke dynamics and speech biometrics to improve authentication security. Specifically, it aims to:

1. Develop a system with keystroke dynamics and speech biometrics, featuring continuous authentication and anomaly-based re-authentication.
2. Gather datasets for authentication purposes, including:
    a. Keystroke dynamics data
    b. Speech biometrics data
3. Identify the most suitable machine learning algorithms for:
    a. Keystroke feature extraction and classification
    b. Speech feature extraction and classification
    c. Anomaly detection for continuous authentication
4. Evaluate the system security performance through vulnerability testing.

SCOPE AND DELIMITATION OF THE STUDY

This study focuses on creating a multimodal biometric authentication system for websites or desktop devices that combines how a person types (keystroke dynamics) and how they speak (speech biometrics) to make digital security stronger. The study will concentrate on the

construction, testing, and design of the system—specifically its accuracy, usability, and security. The procedure includes gathering and constructing datasets for authentication, selecting appropriate machine learning methodologies to categorize typing patterns, extracting features from voice, and identifying anomalous behavior which also entails constructing a functional model of the system that can continually verify a user's identity and prompt for re-authentication if any anomalies are detected. The researchers will develop several testing scenarios to evaluate the system's accuracy, response speed, and efficacy in preventing fraudulent or faked attempts. The study will demonstrate how real-time detection of anomalous activity might initiate re-authentication to ensure session security. It will also examine the system's scalability and its integration into current security frameworks.

This study is limited to just keystroke and speech biometrics since it will not incorporate other kinds of security features or mix in other biometric methods that might enhance security and dependability. The system is also limited to multiple languages or accents; speech biometrics section will only operate with English inputs; the system will only concentrate on typing style and voice characteristics. However, typing analysis for keystroke dynamics will be performed using a given phrase but in a controlled condition that can make the phrases strong and more secure and it won’t look like easy and natural word phrases.

SIGNIFICANCE OF THE STUDY

This study will not only give users a better authentication system, but it will also complement personal data security and serve as an alternative to traditional 2FA systems. This

will instead harness continuous authentication that can help minimize session hijacking as well as unauthorized access.

This study can be of importance to different sectors and industries, such as financial institutions, health care providers, and government agencies, which handle confidential information. This system has a lot to offer in ensuring that these sectors and industries enhance safeguarding and patronage of reduced hijacked sessions and unauthorized access during their operation.

Concerning technology, this multimodal authentication system will enhance the security scope considerably, especially in the field of artificial intelligence and biometrics. Future researchers can use this as a reference in terms of looking for alternative authentication methods.

DEFINITION OF TERMS

- **Biometric Authentication:** A security mechanism that uses an individual's unique physiological or behavioral traits to verify their identity. In this study, biometric authentication refers to mechanisms such as fingerprint, face, iris, and voice recognition, which are considered more secure and convenient than traditional password systems for authentication in this study.

- **Multimodal Biometrics:** The integration of two or more biometric authentication techniques to enhance security, reliability, and accuracy. In this study, multi-modal biometrics join features including keystroke dynamics and speech biometrics, to produce a system that is more reliable because of minimizing the NPPFN and NPFPN.

- **Keystroke Dynamics:** A behavioral biometric authentication technique that analyzes a user's unique typing patterns, including rhythm, speed, pressure, and key press durations. In this study, keystroke dynamics data are processed using machine learning algorithms to distinguish legitimate users from impostors.

- **Speech Biometrics:** A form of biometric authentication that analyzes an individual's voice characteristics, such as pitch, tone, accent, and speaking style, to verify their identity. In this study, we exploit speech biometrics as security alternative to passwords, based on distinctive voice characteristics missed to reproduce.

- **Continuous Authentication:** A security feature which ensures users' identity is legitimated at various instances throughout a session In this research, continuous authentication assesses live user behavior (e.g. typing rhythm and talking patterns), thus minimizing security threats like session hijacking and unauthorized access.

- **Random Forest:** A supervised machine learning algorithm used for classification and pattern recognition through the use of multiple decision trees. In this study, to enhance the accuracy and to mitigate overfitting in keystroke dynamics, Random Forest is used for the analysis and classification of the record data.

- **Anomaly Detection:** A technique used in cybersecurity and machine learning to identify deviations from expected patterns. In this study, anomaly detection is necessary when an unauthorized access attempts to be recognized and raising re-authentication or blocking user, when a user's behavior changes significantly from the normal behavior.

CHAPTER 2

LITERATURE REVIEW

CONCEPTUAL LITERATURE

1. Theoretical Foundations of Multimodal Biometric Authentication

Biometric authentication makes use of the inherent physical and behavioral characteristics of individuals for making an identity verification, hence proving much more secure than traditional methods. Passwords or PINs can be examples of traditional methods. Recent technological advances in the field of biometrics have borne the establishment of multimodal biometric systems which include the fingerprints, the iris, or facial recognition among other traits in the person's identification. Such systems allow for better accuracy in authentication and safety-not only but are proofed more towards spoofing protection and, in addition, flexible enough to portray different users (Figueroa, 2024). Moreover, multimodal complication of biometric fusion in authentication includes data fusion drawn from different traits into an enhanced and reliable mode of confirming identity. Different forms of fusion exist before matching, like sensor-level fusion and feature-level fusion and after matching, like match score, rank, and decision-level fusion, which mainly combine results from individual authentications. The level of fusion chosen determines how much security, accuracy, and efficiency will be increased in the system. (Terfa et al., 2021).

They have set standards for behavioral biometric authentication and multimodal authentication. One of the theoretical models put forward is Fitts' Law, which relates movement time with target distance. It defines the time taken to move a controller, that is, a pointer like a mouse cursor or a finger or a hand, to the target to trigger an action such as clicking, tapping, or

grasping it (Budiu, 2024). This theory is particularly germane to keystroke dynamics and touchscreen-based authentication, being able to analyze speed, accuracy, and consistency of user input. While demonstrating modeling capabilities for user interaction patterns, Fitts' Law also provides insights into the differentiation of legitimate users from imposters.

Another important theoretical framework is Signal Detection Theory (SDT), which evaluates how the performance of biometric systems discriminates between genuine and fraudulent users. It discriminates true sensitivity from the personal biases involved in user responses, leading to improved strategies in decision making and choice (Maniscalco et al., 2024). In this way, SDT allows authentication systems to optimally trade off access control and usability via strategic threshold values, minimizing both false acceptances and rejections. This is particularly useful in speech biometrics whenever small but natural deviations in speech patterns lead to wrong decisions.

Multimodal biometric systems act upon well-established paradigms of how integration of multiple biometric traits would be an advantage in enhancing security and accuracy. Bayesian Decision Theory, which has been quite widely employed, offers a probabilistic model for decision-making. The general idea is to ascertain the best possible outcome by taking into account both past observations and the current relevant contextual factors. It provides an organized way to make the most rational decision based on the data at hand (Gad, 2024). Such an approach is useful in biometric authentication as it refines the matching scores by evaluating the true- and false-match likelihoods associated with the attempt. Acts by continuous learning from the upcoming data enable the Bayesian models to adjust according to the user behavior; thus, an improvement in both security and usability.

Another significant approach is the Fuzzy Logic-Based Model, which particularly addresses uncertainty and variation in the biometric data. In essence, fuzzy logic is therefore an adaptive scheme that performs advanced decision tree processing. This method enables a better fit with rules-based programming, contrasting with the crisp/boolean logic in that it offers degrees of truth beyond true and false. In theory, this method better emulates real-world situations in which approximations of absolute truth or falsehood are very rarely met (Scott, 2023). The model is especially useful in multimodal systems incorporating physiological and behavioral traits, which makes the Fuzzy Logic-Based Model a more flexible approach toward decision-making. Contextual variances could alter a person's speech biometric by slight degrees owing to environmental noise or stress; nonetheless, a fuzzy logic-enabled system may well authenticate them by weighing many acceptable alternative conditions.

One of the newer developments in multimodal biometrics is the Deep Learning-Based Model, which involves the application of neural networks for the automatic learning and pattern extraction of biometric data. Feature extraction and classification using CNNs and RNNs have been successfully applied to pattern and speech recognition (Holdsworth and Scapicchio, 2024). Deep learning models provide better performance to the system as they lessen the manual feature engineering training burden and improve recognition accuracy in conditions that may be challenging. Moreover, these models are improved further with ongoing training on varying datasets, making them more resilient to spoofing attacks and variations.

2. Security Frameworks in Biometric Authentication

Threat Model-Based Security Frameworks is one of the most useful models that can help for protecting the system where the important data of the user and the server were sent and

received. Having this biometric authentication needs a tight security model that can help the user to prevent some different types of risks and possible attacks in the system. In order to prevent attacks, this model categorizes various types of threats and helps prevent them. These types of attacks take advantage of vulnerabilities in the system in order to gain access to it without authorization or to interfere with the way it functions (Nair & Mariappan, 2023). To protect against these problems, multimodal biometric systems use tools like liveness detection and real-time anomaly detection which check for strange behavior or signs that someone might be faking the biometric input.

Based on the concept that no person or device should be trusted automatically, the Zero Trust Architecture (ZTA) is another significant security architecture; rather, it guarantees that every access request is thoroughly examined. Since it requires continual biometric and contextual identification, ZTA considers all users and devices may be dangerous, unlike previous security models that trusted network users (Zohaib et al., 2024). Continuous verification helps secure remote access whether employees use VPNs from home or corporations utilize cloud services. These security models strengthen and improve dependability in a multimodal biometric system by better complying with privacy standards and protecting it from attacks and vulnerabilities that previous authentication methods failed to manage.

3. User-Centered Design and Usability Theories

While biometric authentication is primarily about security, it also has to be user-friendly. A system that frustrates users will struggle with adoption, no matter how secure it is. Human-Computer Interaction (HCI) Theory highlights the importance of creating, developing, and assessing interactive interfaces to improve user interactions with computing devices (Kanade, 2024). Keystroke dynamics and speech biometrics work well because they don’t

require extra effort, users just type or speak as they normally would. However, factors like background noise, different devices, or even a user's mood can affect how well the system recognizes them.

To make biometric authentication better, usability principles like Nielsen's Usability Heuristics come into play. These principles emphasize and determine how easy and intuitive a product is to use, ensuring recognizable elements and an ideal user experience (Zelezny, 2024). For example, if a voice authentication attempt fails, the system could prompt the user to adjust their tone instead of simply rejecting them outright. Additionally, user acceptance is key. The Technology Acceptance Model (TAM) explains that when a technology is perceived as both useful and easy to use, individuals are more likely to adopt and integrate it into their activities (James and Bewsell, 2024). While behavioral biometrics can offer a frictionless experience, privacy concerns can make users hesitant to adopt them. Being transparent about data security, allowing users to opt in or out, and providing clear privacy policies can help build trust in multimodal biometric authentication.

RELATED LITERATURE

1. Biometric Authentication Systems

Biometric authentication systems utilize unique physiological or behavioral traits of individuals for identification or verification, offering a higher level of security compared to traditional methods such as passwords and PINs. These systems can be broadly categorized into physiological biometrics, which includes fingerprint recognition, face recognition, hand geometry, and retina scans, and behavioral biometrics, such as keyboard dynamics and voice

recognition (Lajçi & Misini , 2022). Biometric authentication is increasingly integrated into various sectors, from financial services to healthcare and government domains, due to its ability to provide strong, user-friendly security (Basare et al., 2023). The primary advantage of biometrics lies in the difficulty of duplicating or stealing biometric data, as each individual's biological or behavioral features are inherently unique. Specifically, behavioral biometric methods are becoming increasingly popular as they can be used to authenticate users without the need for physical contact (Kamble et al., 2024).

Behavioral biometrics analyzes how you act online, how you interact with your phone, and how you type. It more focuses on patterns of behavior like keystroke or how a phone is held, unlike physical biometrics such as fingerprint or face, it enhances the prevention of identity theft. Behavioral biometric systems cannot be fooled using stolen data as authentication happens dynamically as it provides increased convenience and security (Sharma & Elmiligi, 2022). Unlike other authentication methods, behavioral biometric authentication does not need the user to perform any unnecessary actions to authenticate. However, a study also stated that this kind of authentication generates a large dataset that slows down the authentication process and its reliance on the user to interact with the device (Finnegan et al., 2024). As behavioral biometrics popular nowadays, it will become more relevant in the future (Sharma & Elmiligi, 2022).

2. Keystroke Dynamics in Authentication

Keystroke dynamics is an authentication method that authenticates an individual based on his/her typing pattern. Keystroke dynamics can be used either for authorizing users on a system, or for detecting intrusions either in offline or online scenarios. After measuring the timing behavior of a user while interacting with the keyboard, a proper algorithm is selected for user

verification (Najwa Altwaijry, 2023). Moreover, since it relies on only the keyboard, keystroke dynamics is an inexpensive and non-intrusive method.(Shadman et al., 2023). In such a context of personal data security, keystroke dynamics can avoid fraud by validating the user's authentication credentials every time they access the system to use it.

Furthermore, the typing pattern exhibited by individuals is a major variable when it comes to making keystroke dynamics effective for user's identification. Features such as key durations and latencies, transformed into image-like structures, have been shown to allow for accurate identification using machine learning algorithms like CNN and Random Forest. In one study, Random Forest attained significantly higher performance than teh CNN and achieved 93% accuracy rate, which can be seen that timing-based features are highly applicable for realistic and continuous authentication scenarios (Sharma et al., 2023).

Further studies have developed other features and algorithms to enhance the accuracy and robustness of keystroke dynamics-based authentication systems. One such innovation is the Distance Enhanced Flight Time (DEFT) feature set, which combines flight times with physical key distances, making it device-agnostic. This approach achieved an accuracy rate greater than 99% with an Equal Error Rate (EER) of less than 10%, which means that keystroke dynamics can perform well across multiple devices and contexts, thereby adding an extra layer of security from threats/security attacks (Kaluarachchi et al., 2023). This demonstrates the robustness of keystroke dynamics to be secure and unique no matter what device is utilized.

The reliability of keystroke dynamics as a unique biometric feature was further affirmed by a comparative analysis of various keystroke-based authentication methods. The study emphasized the importance of feature selection and model choice in achieving high accuracy. It

was found that timing features such as hold time and down-down time significantly improved performance, making keystroke dynamics a viable option for secure user authentication (Amin & Di Iorio, 2025). These findings underline the inherent uniqueness of typing patterns, which are resistant to spoofing and mimicry by attackers.

Machine learning techniques have as well enhanced the precision and security of keystroke dynamics-based systems. One study applied Transformer-based models to mobile keystroke biometrics, significantly outperforming previous methods with an Equal Error Rate of just 3.25%. This work emphasized the challenge of mobile keystroke biometrics and demonstrated how advanced algorithms such as the TypeFormer can bridge performance gaps in free-text scenarios, which are particularly vulnerable to attack (Lee et al., 2022). The use of complex models such as Transformers highlights the difficulty in replicating an individual's typing behavior, making attacks on such systems more time-consuming and challenging.

Additionally, multiple biometric modalities has been proven as an approach to improve the robustness of keystroke dynamics systems. One research paper fused EEG signals and keystroke dynamics, which can achieve a good identification performance 99.9%. In this multimodal approach, the combination of the multiple biometric's factors (i.e., haemodynamic) was shown to offer an enhanced tolerance against hackable attempts to make that mimicry or impersonation does not possible over short period of time (Rahman et al., 2021). The evidence provided for both of them, by combining several sources of biometric, enhances the case of the long term security of a biometric based on keystroke.

The continued development of machine learning techniques and feature extraction algorithms has continued to enhance the performance and security of the KD based

authentication systems. One of these approaches was the use of a fine-tuned Convolutional Neural Network (CNN) model with synthesizing data techniques, and it achieved 99.95% accuracy. This approach suppressed abnormalities, optimized feature distribution, increased the robustness of the system and was not easy to attack. Such a complicated approach suggests that the keystroke dynamics can be a viable alternative to user authentication for the long term, making it less prone to immediate hacking (AbdelRaouf et al., 2023).

2.1 Keystroke Data Collection

Data collection in keystroke dynamics means collecting several temporal features that mainly focus on timing. These features range from key duration/hold time, flight time, latency time, digraph and trigraph time, n-graph time, to both total and average time (Roy et al., 2022). Specifically, latency time and hold time are the two main timing features which are analyzed in keystroke dynamics. The latency time presents the time interval between pressing two keys (a key press/release and the next following key press/release), while hold or dwell time measures how long a key is held down before being released. This means that between two succeeding keystrokes, six timing features can be evaluated including the (1) hold time of two keys, (2) time between key presses, (3) time between key releases, (4) time between first key press and second key release, (5) time between first key release and (6) second key press (Amin & Iorio, 2025). Since typing behavior is observed to be specific to every user, the analysis and usage of these timing features can further help in thoroughly verifying users based on their regular keyboard use. Such a method increases security and decreases the chances of unauthorized access; which is a beneficial element of multimodal biometric verifications, especially when combined with other traits like speech recognition.

Furthermore, these data can be gathered using dedicated software or custom-designed logging systems that monitor a user's typing behavior in real time. Input devices like keyboards and touchscreens often include sensors that capture the exact timing and sequence of keystrokes (Roy et al., 2022). Its strength as a good authentication method lies in how unobtrusive it works behind, its ability to operate with minimal user involvement.

2.2 Keystroke Feature Extraction and Classification Algorithms

When looking at keystroke dynamics data, feature extraction is a very important step as it helps sort and understand the data better which finds trends in how a person types that don't change over time. People often measure digraphs, which is the time between pressing two keys, trigraphs, which is the time between pressing the first and third keys, and n-graphs, which is the time between pressing the first and last key in a set. The hold time (how long a key is held down) and movement time (how long it takes to move from one key to another) are also important (Yaacob et al., 2020). The patterns are then gathered, the data is handled, and the most important parts are cut down to make the system run faster and more correctly. Feature extraction is important because it helps the system focus on the most useful and regular parts of how a person types, which makes it easier for the system to recognize users.

Machine learning is used to sort the typing data and confirm a user's identity once the important traits have been found and the data has been collected. In keystroke dynamics, there are a number of machine learning methods that are used. Decision Trees (DT) are one example—they are easy to understand and use simple rules to classify users based on how they type, but they can sometimes overfit the data, so they need adjustments like pruning to work better (Wang & Hou, 2024). Another method is Random Forest, which combines several

decision trees to make more accurate predictions and is known for being both reliable and strong in different tasks. Support Vector Machines (SVMs) are also used—they separate different types of data using a line or plane, and an adaptive version of SVM can improve accuracy by adjusting to changes in someone's typing without needing a full retrain (Shadman et al., 2023). Neural Networks, especially deep learning models, are also popular because they can learn very detailed patterns from large amounts of data, helping the system become more accurate over time (AbdelRaouf et al., 2023). The best algorithm to use depends on what the system needs—some are faster, while others are more accurate or better for complex data.

3. Speech Biometrics in Authentication Systems

Speech biometrics refers to the pattern of voice frequencies and other speech-related features used to identify users. The human voice constitutes behavioral biometrics, which manifests in a unique and specific way that varies between individuals according to their characteristics of frequency, loudness, timbre, accent, rhythm, or tone. The combination of vocal apparatus and behavioral patterns determines these vocal characteristics, which differ from person to person (Zeynep Örpek et al., 2024). Such high accuracy in authentication is achieved by voice biometrics due to the interplay of physical aspects like structure and behavioral aspects like a pattern. Other factors affecting voice recognition precision include background noise and emotional or health state; this necessitates the development of other strong algorithms for the improved reliability of the system.

In the voice biometrics field, speaker authentication technologies are utilized as standard reference monitors and include Automatic Speaker Verification (SV) and Verbal Information Verification (VIV) (Ibrahim, 2021). Speaker Verification (SV) ascertains that a claimed identity

corresponds to the stored voiceprint, while Verbal Information Verification (VIV) checks the spoken information for validity. These techniques have been widely used in banking authentication, call centers, and secure access systems, largely improving identity verification. The possibilities for VIV are enhanced by the ever-growing reliability of NLP, which also distinguishes genuine and fraudulent interactions.

3.1 Speech Data Collection

Speech data collection in terms of biometric authentication are categorized into two methods which are the text-dependent and the text-independent systems. In text-dependent systems, users are required and prompted to speak a given specific phrase or passphrase, which enables for a more controlled and stable environment for data collection (Wells, 2021). Additionally, this approach typically offers higher accuracy because the speech content remains the same across different sessions. On the other hand, text-independent systems allow users to speak freely, without having to look and speak a predefined phrase (Wells, 2021). Although it is given that this provides a more natural authentication process, it is observed to present higher unpredictability in the speech data, a challenge that makes feature extraction and classification harder. In both cases, modern speech recognition systems use microphones or other audio sensors to capture the user's voice, which is then processed and to be analyzed further. Once such speech signals are captured, several acoustic features are extracted to represent the unique vocal characteristics of the speaker. These features then serve as the input for machine learning or deep learning models used in speaker verification and identification systems.

3.2 Speech Feature Extraction and Classification Algorithms

In speech biometrics, the primary task is to extract relevant features from the voice signal that can uniquely identify a speaker. Mel-frequency cepstral coefficients (MFCC) are among the most commonly used features for speech processing, capturing the spectral properties of the audio signal and mimicking the way humans perceive sound. MFCCs are widely used as key features in speech signal analysis, particularly in speech recognition, relying on the cepstral and Mel scales. They are known for their effectiveness and ease of use in processing speech (Abdusalomov et al., 2022). The MFCC features are extracted from recorded speech signals using phonogram and spectrum transformation techniques. Other important features include pitch, which indicates the perceived frequency of the voice, and tone, which can vary between individuals. Linear Predictive Coding (LPC) is another popular method, and unlike MFCC, LPC mimics the vocal tract's structure when producing sound. It analyzes speech by dividing the input signal into frames and applying windowing to minimize discontinuities at frame boundaries (Labied & Belangour, 2021). These features are then classified using various machine learning algorithms. The goal of feature extraction is to select the most distinctive and strong features that accurately represent an individual’s voice while minimizing variability due to background noise or other external factors.

Once speech features are extracted, machine learning algorithms are used for classification to authenticate or verify users. Hidden Markov Models (HMMs) is one of the most effective classifiers for speech recognition and is widely used for this purpose. Its flexibility allows it to adapt to different structures, making it easy to train, implement, and efficiently apply in various speech-related tasks (Malik et al., 2020). Deep learning techniques, such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), are as well

widely used in speech processing and have shown great success in tasks like speech recognition, speaker identification, and speech synthesis (Mehrish et al., 2023). The choice of classification algorithm depends on the available data, system requirements, and the desired trade-offs between accuracy, speed, and computational complexity.

4. Multimodal Biometric Authentication Systems

Multimodal biometric authentication systems improve security by combining various biometric modalities. Such technique then makes up for the shortcomings of unimodal techniques as using multimodal frameworks rather than unimodal biometrics eliminates authentication constraints. Traditional biometric systems use a single modality, such as fingerprints, facial recognition, or iris scans, but they are susceptible to spoof attacks and manipulation, inter-class similarity, intra-class variance, non-universality, and noisy data (Sumalatha et al., 2024). Multimodal authentication improves the accuracy and security by combining multiple biometric attributes. The combination of physiological and behavioral biometrics creates a more thorough verification framework, making multimodal systems a practical alternative to traditional two-factor authentication.

Fingerprint & face, fingerprint & iris, and face & voice are some examples of combinations that use unique biometric properties to achieve an appropriate balance between accessibility and performance. In the study that evaluated the aforementioned combinations, it is concluded that voice-based multimodal combinations improve accessibility and ease of use, making them ideal for mobile and remote authentication (Maiti & Basak, 2024). In addition to this, another known substitute for conventional authentication techniques is keystroke dynamics, which are further improved by machine learning, deep learning, and neural networks. The

probability of keystroke dynamics to heighten security and replace password-based systems is supported by studies that demonstrate that Random Forest performs best for machine learning, CNN-GRU-DBN excels in deep learning, and Bi-CNN excels in neural networks (Arsh et al., 2024).

Keystroke dynamics and voice biometrics offer a promising combination for multimodal authentication, utilizing unique behavioral characteristics for increased security. Multibiometric systems can be constructed in either serial or parallel method, depending on the application scenario and performance requirements. Serial processing is simpler and much more straightforward to implement since biometric features are analyzed consecutively, although this process is reliant and dependent on the performance of the first modality. Parallel processing, on the other hand, allows multiple biometric inputs to work independently, with increased overall performance by feature, score, or decision level fusion (Poornima & Subramanian, 2022). Fusion further increases accuracy and security by maximizing complementary features, such that if the performance of one modality decreases temporarily due to reasons like fatigue or sickness, the other can cover. Comparatively, this improves security as it is harder for attackers to simulate numerous modalities simultaneously, along with ensuring authentication is robust despite variations in personal characteristics.

Moreover, RNNs are powerful for anomaly detection of time series data, and particular architectures such as LSTM-RNN or stacked RNN are appropriate depending on the application. Some uses of RNNs are mouse movement, keystroke, handwritten password, and palm print authentication, which provide greater security by being more difficult to impersonate (Ackerson et al., 2021). Through the usage of deep learning algorithms like Long Short-Term Memory

(LSTM), authentication systems can learn and adapt to users' behaviors dynamically over a period of time, diminishing false positives and enhancing security.

5. Multimodal Fusion Techniques

Multimodal biometric systems merge several forms of fusion (for instance, feature level, score level, decision level) with an objective of strengthening the accuracy, security, and robustness. Feature-level fusion refers to the cases of combining different feature vectors from the same biometric category. This area of investigation is limited since different biometric modalities are incompatible with one another in their features, but it is expected that they would improve recognition performance better, since the features are richer and give more meaning (Sumalatha et al., 2024). On the other hand, score-level fusion generates a final score for the identification/authentication by combining the match scores from different biometric modalities, which researchers favor on account of its simplicity relative to the other two fusion types (Sumalatha et al., 2024). Decision-level fusion is the process of arriving at the final decision by combining decision outputs. Even though it offers high accuracy, it is more robust to errors and flexible to heterogeneous data types, decision-level fusion is less often used in multi-modal recognition systems because it uses less data (Sumalatha et al., 2024). Thus, decision-level fusion is less suitable than score-level fusion (Safavipour et al., 2022).

6. Network Overhead in Multimodal Biometric Authentication

Network Overhead refers to additional computational and communication resources required to transfer and process biometric data across systems resulting in an increase in CPU load, memory usage, and bandwidth consumption, encompassing overhead in multimodal biometric authentication (Mohamed et al., 2025). In systems combining both keystroke dynamics

and speech biometrics, overhead originates from two main sources, namely static attributable to packet headers, encryption and authentication metadata and dynamic as retransmissions, buffering delays and jitter correction. Of course, controlling these is critical for the security, efficiency, and responsiveness of the authentication session. Network overhead is also aggravated when keystroke and speech biometrics are combined into one system because there is increased complexity in the amount of data sent through the network (Feng et al., 2021). The processing and encryption of multiple biometric signals also boosts the overhead in packets. Fulfilling it also means that with low latency and a real-time performance standard, the dynamic overhead caused by retransmissions, jitter, and buffering must be efficiently managed (Feng et al., 2021).

In fact, speech features such as Mel-frequency cepstral coefficients (MFCCs) or deep learning-based embeddings raise packet sizes, thus making real-time authentication quite difficult (Feng et al., 2021). So, re-transmission makes up for unreliability of the data but comes with additional latency and thus may also interfere with the usually swift nature of authentication processes. Encryption is necessary to secure keystroke and speech data but entails other overheads in terms of computation and transmission. This is exceedingly pertinent in multimodal systems of biometrics, since each type will bear its own security and processing requirements (Feng et al., 2021). Thus, it involves reducing both static and dynamic network overhead, to achieve fast, secure, and accurate user verification.

Static overhead is mainly from headers of packets, encryption, and the metadata for authentication. In contrast to generally accepted practices of keeping latency low, randomized TCP segmentation purposely adds header overhead to prevent possible channel attacks (Alyami et al., 2023). This applies in the case of multimodal systems, whereby the encryption of

biometric data such as keystroke timings and speech features literally increases packet sizes and processing demands, hence increases resource costs (Alyami et al., 2023). Thus has been proposed a framework of short packets for communication, based on which a balance between security and throughput is achieved; therefore, this serves as an answer for multimodal authentication systems concerned with speed and protection simultaneously (Feng et al., 2021). Data integrity maintained through retransmissions and jitter corrections generate dynamic overhead. Banerjee et al. (2023) investigated whether old packets should be retransmitted or new packets prioritized to support freshness without delay, which is important for continuous authentication systems that are timing sensitive. It was revealed by Wu et al. (2022), who studied sliding network coding (SNC), that while such coding does provide reliable transmission in wireless networks, it also incurs delay, which could be detrimental to applications such as speech recognition that are time-sensitive (Wu et al., 2022).

Buffering and jitter correction have their contributions to multimodal systems with real-time capabilities. Research on 5G networks highlights the importance of jitter control, which varies with delay packet arrival, for smooth and secure data transfers, particularly in time-critical applications (Rodriguez-Martin et al., 2025). With biometric authentication, jitter might interfere with the temporal signaling of voice signals, therefore reducing recognition accuracy. Buffering may help smooth out some of these variations but introduces latency of its own. The fine balance between buffering and responsiveness becomes difficult when seeking to synchronize speech and keystroke data for real-time analysis (Rodriguez-Martin et al., 2025). Therefore, a good adjustment of jitter and buffering control should guarantee a good compromise toward system accuracy and user experience.

7. Security Threats and Countermeasures in Multimodal Biometric Systems

ML models need high-quality, reliable data to work it right, and to get that data you need to use and get it from dependable sources (Alhomayani & Mahoor, 2020). These ML models are used by biometric systems to identify and prove people based on things that are different about them, like in this multimodal of how they speak or type. Making phony biometric samples that look like genuine users or creating new ones is a frequent way to hack these systems and deceive the models into making incorrect decisions (Siddiqui et al., 2022). To prevent data poisoning, ensure safe training data by removing harmful samples through validation, anomaly detection, and data cleaning. Ghilom and Latifi (2024) recommend regularly monitoring training data for unusual patterns or bias that may indicate an attack.

It is also important to use biometric security and safe data keeping for the multimodal biometric identification system to stay safe and secure. Biometric encryption converts raw user data into protected forms that are hard to access or abuse (Zahoor, 2023) to keep biometric templates safe while they're being sent or kept. This means biometric data is never saved in its original form—instead, it’s converted into secure patterns used only for matching during login. Some encryption methods even allow the system to work with encrypted data directly, without needing to decrypt it, which helps protect privacy even more.

To make authentication smooth across different devices and places, a flexible storage setup is needed. Storing data locally offers quick and private logins without needing internet, but it limits access to one device. Cloud storage allows users to log in from anywhere and keeps backups, but it needs strong security to avoid hacking. A good solution is a hybrid model that

uses both—some data is kept on the device for fast access, while other parts are safely stored in the cloud for multi- device use (Zimik & Keishing, 2022). To ensure no one tampers with the biometric data, the system can use integrity checks, encrypted communication, access logs, and real-time monitoring to spot and stop suspicious activity. By combining these tools, the system stays secure, user- friendly, and protects sensitive biometric details from being misused.

8. Continuous Authentication and Anomaly Detection

The growing complexity of cyberattacks requires authentication mechanisms that offer persistent security. Continuous authentication that authenticates a user in real time based on behavioral characteristics has become a key solution in this context. For example, a security technique known as Behavioral Biometric Continuous Authentication increases the primary login process with constant monitoring of user activity to validate identity. This is based on several behavioral attributes, such as hand movement, keystroke dynamics, gait, touch gestures, user profile, and power consumption. Behavioral biometrics can also be divided into key categories such as gait, motion, speech, keystroke dynamics, and gestures, and the use of more than one modality can strengthen authentication confidence (Skalkos et al., 2021). Unlike traditional authentication methods, continuous authentication continuously attempts to detect anomalies or deviations and prevent unauthorized access. This is usually achieved through the combination of disparate biometric modalities and contextual information.

Machine learning algorithms also have an important role to play in this anomaly detection process. ML algorithms such as K-NNs and SVMs can be used to aid in the detection of deviation from a user's pre-set baseline and hence makes them very valuable for continuous authentication systems (Natha, 2022). For instance, K-NN determines the degree at which the

current behavior is close to that of the past, thus detecting outliers that could represent suspicious activity. SVM, particularly in one-class classification mode, has been shown to work very well at distinguishing anomalies from normal behavior, offering a robust mechanism to detect fraudulent activity based on biometric information. Anomaly-based re-authentication is also the core element of continuous authentication systems. It seeks to detect unusual patterns in the user's biometric information or contextual data. Where there is an anomaly, the system triggers a re-authentication session to verify the identity of the user. Anomaly detection is able to comprehend behavioral information so that it can detect inconsistencies which may indicate fraud or unauthorized access (Zhang et al., 2021). Contextual information includes environmental and situational data which can provide hints regarding the legitimacy of the current session.

Information acquired from context-aware sources is harder for attackers to manipulate, lowering the rate of successful spoofing attacks (Progonov et al., 2022). For example, if a user normally accesses a system from a particular location and their current location is marked as outside their usual area, the system can identify this as an anomaly. In the same way, context data can include device information, time of access, or type of network connection utilized. Enhanced security monitoring on multiple devices and remote locations is facilitated by the flexibility of continuous authentication. For cloud and VPN-based authentication systems, this is particularly relevant as traditional approaches might not detect session hijacking or unauthorized usage.

9. Security Testing and Evaluation

The integration of security testing approaches in the design and development of web-based biometric authentication systems plays an important role in providing secure and

reliable services to support such systems against different cyber attacks. It is critical to support the development of web-based biometrics authentication systems with security testing approaches as such systems will face a range of security threats. Some researchers emphasize the importance of authentication testing, which assesses the accuracy with which computer systems can reliably confirm the identities of users based on biometric measures including voice and facial recognition. Another research implemented a voce biometric login into a web application augmented with a modified GMM to support for improving the success rate, as well as the security features are integrated into the biometric authentication pipeline (Kamiński et al., 2023). Another study assessed the robustness of voice-based authentication in the presence of simulated adversarial attacks including replayed response attacks and deepfake audio, and unveiled failure cases at the system deployment level where a number of defense policies were not applied (Dziegielewska et al., 2021).

In addition to these mechanisms, penetration testing and threat modeling are also key approaches for discovering and mitigating security issues. An extensive analysis of WebAuthn and FIDO2-based systems analyzed how biometric authenticators worked across different browsers and operating systems, which assisted developers in knowing the weak points of a system, and how to design a system with better cross-platform security (Rivera-Dourado et al., 2021).

Moreover, vulnerability assessments and source code review during development can identify security weaknesses prior to deployment. Recent works provide the comparative summary from different biometric traits as a selection for verifying, and analyses the crucial performance measures such as False Acceptance Rate (FAR) and False Rejection Rate (FRR) to

determine, which modality combinations have the best trade-offs between security and usability (Alrawili et al., 2024).

Also, statistical analysis techniques may apply in detecting attacks on biometric systems. One such method was able to identify almost all statistical inference attacks, and highlighted the importance of introduction of sound evaluation methods as early as in development rather than relying on post-deployment fixes (Habib et al., 2022).

SYNTHESIS

Digital threats bring about the need for secure and user-friendly modes of authentication. Though traditional two-factor authentication (2FA) is still in vogue, many cyberattacks targeted at penetrating the same have been reported. This ATB has led researchers to expand the study of applying a multimodal system for biometric authentication, employing a mixture of human traits to improve security and usability functions (Figueroa, 2024). While unimodal systems work with a single biometric input, a multimodal system mixes different biometric inputs together, making it more resistant to spoofing.

This study focuses on keystroke dynamics and speech biometrics, two passive means of behavioral user authentication. Keying and speaking are natural to the same level, making the methods realistic to use with such a system in real scenarios. Models such as Fitts' Law, discussing motion by humans inside interfaces (Budiu, 2024) and Signal Detection Theory, which would discuss the trade-off between false acceptances and rejections (Maniscalco et al., 2024) are strongholds of these methods. In such environments, Bayesian Decision Theory

assisted the systems in learning from previous data (Gad, 2024) while fuzzy logic improved the flexibility of those systems under perturbation, such as background noise or stress (Scott, 2023). With further developments, deep learning has considerably improved biometric performance. Such models as CNNs and RNNs help to identify complex patterns, thereby bettering system accuracy under noisy or variable operating environments (Holdsworth & Scapicchio, 2024).

Keystroke dynamics identifies users through the analysis of typing behavior, which includes parameters such as key hold durations and interval times between keystrokes. These patterns are unique, can easily be captured using standard keyboards, and cost a dime. Machine learning models like Random Forest achieved a whopping 93% accuracy on the classification task (Sharma et al., 2023). The method Distance Enhanced Flight Time (DEFT) incorporates the layout of the keyboard, improving the consistency of measurements across different devices (Kaluarachchi et al., 2023). In addition, keystroke data accomplishes continuous authentication, thus maintaining security throughout an entire session (Amin & Iorio 2025).

Speech biometrics analyses those vocal characteristics that distinctly identify an individual, including pitch, tone, and rhythm, which arise from a complex interplay of physical and behavioral attributes (Zeynep Örpek et al. 2024). The practical applications of voice recognition face challenges emanating from background noise or the recognition subject being unwell, while methods such as MFCCs and LPC fairly extract consistent features (Abdusalomov et al. 2022; Labied & Belangour 2021). Model approaches such as HMMs and neural networks provide reliable speaker recognition even when the content varies (Malik et al. 2020; Mehrish et al. 2023).

Merging keystroke and speech biometrics fortifies the assurance of the system. If one characteristic is compromised, the other will compensate. The score-level and decision-level fusion methods bring both the traits together so that performance may be enhanced (Sumalatha et al., 2024; Safavipour et al., 2022). Studies on such systems have shown penetration to the challenges of the real environment, such as spoofing, noise, or device variability (Maiti & Basak, 2024). Hybrid deep learning models like CNN-GRU-DBN and Bi-CNN have further upsurge the recognition rates (Arsh et al., 2024). The security aspect is one in which users adopt, but usability comes with it. Principles of HCI raise a need for simple, easy systems to be helpful, according to the Technology Acceptance Model (TAM) (Kanade, 2024; James & Bewsell, 2024). Keystroke and speech systems are intuitive because they reflect users in everyday lives as well as speak or sense their inputs. Features such as feedback that seeks to enhance voice input instead of outright rejection improve the user experience (Zelezny, 2024).

Real-time systems tend to both voice and keystroke processing from mobile. Carrying out the process, such systems require high bandwidth and computing power, which would be causing latency due to different reasons (Feng et al., 2021). Some solutions exist and support smaller packet encryption and jitter buffering are there, keeping up performance (Alyami et al., 2023; Rodriguez-Martin et al., 2025).

Aside from the mentioned before, some major areas of concern are spoofing, replay attacks, and data poisoning. Some of the mitigations include encryption, anomaly detection, and secure hybrid storage between the cloud and local servers (Ghilom & Latifi, 2024; Zahoor, 2023; Zimik & Keishing, 2022). Further, models like LSTM, SVM, and K-NN may also be useful in detecting less-crying behavioral changes associated with intrusions (Natha, 2022).

Testing and validation ensure that biometric systems can stand up to their reliability. Biometric systems are examined when under attacks, such as replay voice or impersonation via deepfakes, and strong models and defenses have been used effectively in countering these (Dziegielewska et al., 2021). Improved GMMs and tests compatible with WebAuthn/FIDO2 have also shown promise across browsers and devices (Kamiński et al., 2023; Rivera-Dourado et al., 2021). Tools like statistical analysis and vulnerability testing are useful in exploiting system weaknesses (Habib et al., 2022).

The incorporation of keystroke dynamics with speech biometrics provides much stronger authentication that is much more user-friendly. Unlike traditional systems, it works passively when a user is normally interacting with the system: both methods create an environment for continuous protection and convenience. It comes with machine learning and a well thought out system design that meets modern challenges in cybersecurity whilst retaining user-friendliness. Like much of the suffering advances being made in biometric technologies, such solutions are expected to become a key fixture of developments in digitally much more secure authentication.

CHAPTER 3

METHODOLOGY

MULTIPLE DESIGN CONSTRAINTS, TRADE-OFFS, AND STANDARDS

Multiple Design Constraints and Trade-Offs

Design 1: Sequential Authentication

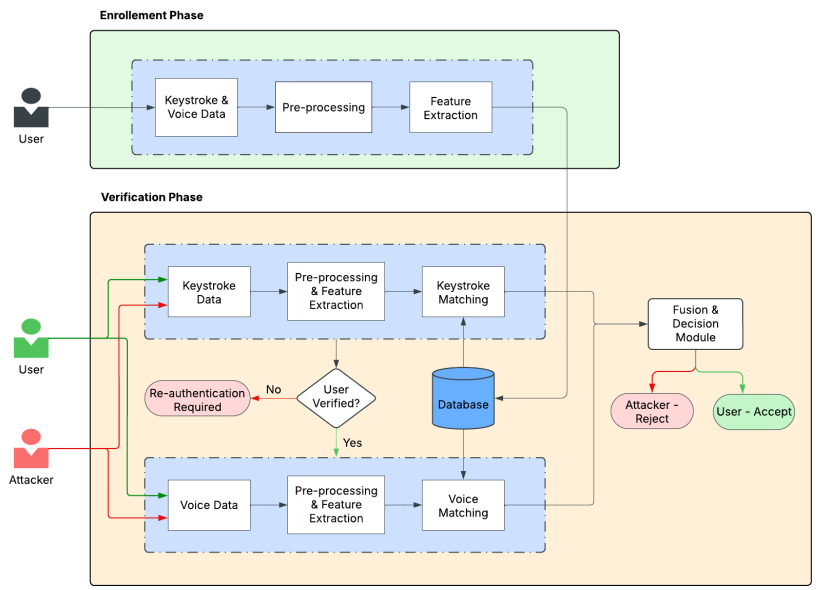

Figure 1. Sequential Authentication Design

Figure 1 illustrates a sequential multimodal biometric verification system incorporating keystroke dynamics and speech biometrics. In the enrollment phase, keystroke and voice samples are collected, then pre-processed and feature extracted, stored in a database. The verification phase then starts when a user attempts authentication by providing keystroke and voice data. Keystroke matching is done first and if it succeeds, voice data will be processed next and both

results are combined to take a final decision. If authentication fails, re-authentication is required or access is denied for attackers.

Design 2: Simultaneous Authentication

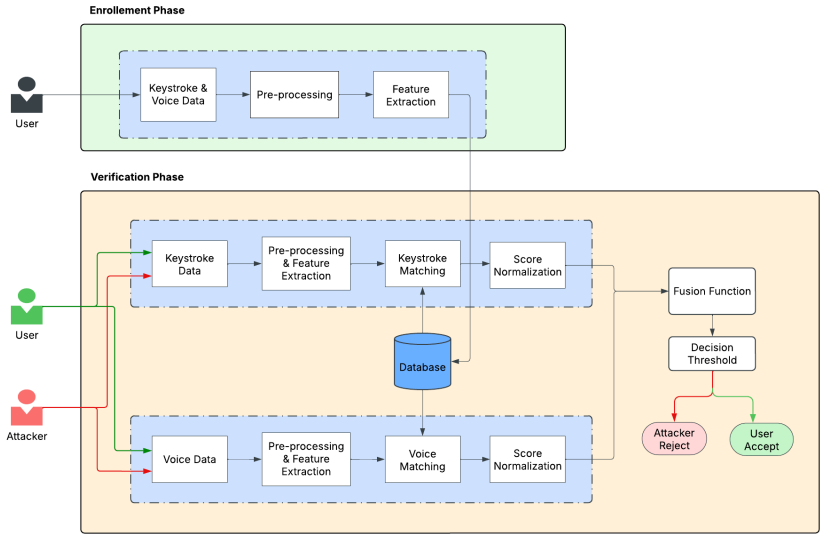

Figure 2. Simultaneous Authentication Design

Figure 2 presents a Simultaneous Authentication Design using keystroke dynamics and speech biometrics as a substitute for standard two-factor authentication. The user starts the authentication process by typing and saying the same passphrase simultaneously. When the user inputs the passphrase, the system records two biometric modalities in real-time: keystroke dynamics, which inspects typing speed, pressure, and rhythm, and speech biometrics, which inspects vocal features. These inputs are then processed through independent feature extraction modules, where distinctive biometric patterns are extracted. The features extracted are passed to standalone verification models, where they are matched against the user's pre-registered biometric templates. The authentication outcomes from both modalities are then fused in a fusion

module, which decides whether access is granted or not based on a predetermined security threshold. When both biometric parameters are successfully authenticated, the system provides access. In case the authentication is unsuccessful, the system can request re-authentication or invoke other security measures.

Design 3: Progressive Authentication

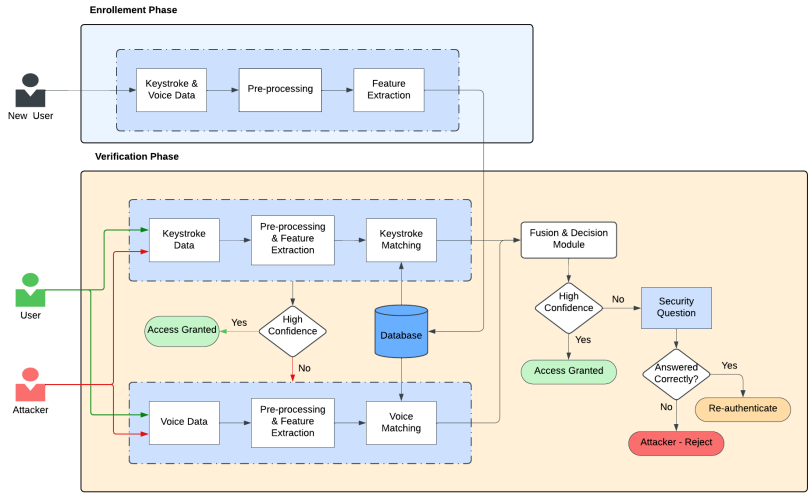

Figure 3. Progressive Authentication Design

Figure 3 shows the progressive multimodal biometric authentication using keystroke dynamics and speech recognition. The enrollment phase shows the collection, pre-processing, and feature extraction of the collected keystroke and voice data then storing it to the database for verification phase. In the Verification Phase, the user or attacker inputs keystroke and voice data, which undergo pre-processing and feature extraction, matching the data stored in the database. The first keystroke, if the process is successful then the access is granted and if not then proceeds to voice data. Next is voice data, same as keystroke, if the process is successful then access is granted. However, if the process is unsuccessful then the system will ask a security question to

the attacker or user. If answered correctly it prompt the user for re-authentication and if not, then the user or attacker is rejected and flagged as the attacker.

Table 1. Comparison of System Design

	Design 1	**Design 2**	**Design 3**
Authentication	Sequential	Simultaneous	Progressive
Biometric Modalities	Keystroke dynamics and voice recognition		
Enrollment Phase	Keystroke and voice data collection and feature extraction		
Verification Phase	Keystroke matching first, then voice matching	Keystroke and voice matching simultaneously	Keystroke matching first, then voice matching, then security question
Fusion of Results	Fusion of keystroke and voice results at the end	Fusion of both biometric results after real-time processing	Fusion of keystroke and voice results after individual matching
Security Measure	Access is granted if both keystroke and voice match	Access is granted only if both keystroke and voice are authenticated	Access is granted if keystroke and voice match; otherwise, security question
Failure Handling	Re-authentication or access denial after failure	Re-authentication or security measures invoked after failure	Security question for re-authentication if both fail
Real-Time Processing	No real-time requirement; matches sequentially	Requires real-time simultaneous processing of keystroke and voice	No real-time requirement; processes sequentially
Complexity	Low to moderate (simple sequential flow)	High (requires real-time processing and fusion of two biometric types)	Moderate to high (sequential steps with added security question)
Attack Resistance	Moderate (attackers may bypass if one modality is compromised)	High (both modalities must be matched simultaneously)	High (additional security question after failed authentication)

Table 1 shows the comparative analysis of three multimodal biometric authentication designs in terms of biometric modalities, authentication, enrollment and verification process, and failure handling. In Design 1, authentication is sequential with keystroke data authenticated first

before voice data and access is allowed only when both are found to be matching. This design is computationally inexpensive, does not involve real-time processing, but only offers fair protection against attacks. Design 2 requires that both biometric traits to be simultaneously verified and the result of the verification should be in real time but it offers high security level and higher resistance against attack but the system complexity is high and real-time processing control unit have to be used to process the result. Design 3 starts a progressive process where the system first transmits some keystroke and some voice data, and finally the user is asked a security question. This pattern does not rely on real time and provides fall back authentication, which brings moderately high complexity and high resistance against attacks. In summary, all designs represent a tradeoff between security, complexity and usability D1 can be described as simple, D2 as secure, and D3 as a best effort to balance both with additional fallbacks.

Table 2. Rubrics for Multiple Design Constraints and Trade-offs

Constraints	1	2	3	4	5
Global	Limited accessibility, struggles with diverse users.	Partially accessible, usability issues remain.	Moderately accessible, adaptable in some cases.	Highly accessible, handles diverse inputs.	Fully accessible with fallback mechanisms
Economical	High computational cost, inefficient processing.	Some optimization but still costly.	Balanced cost, moderate resource use.	Cost-efficient, reduces redundancy.	Highly optimized, minimal processing waste.
Environmental	High energy consumption, redundant operations.	Partially optimized but still power-intensive.	Moderate efficiency with some optimi-zations.	Energy-efficient, reduces unnecessary processing.	Highly optimized for minimal energy use.
Societal	Inconvenient, time-consuming authentication.	Some usability but still requires effort.	Balanced security and convenience.	Seamless authenticati on with fallback options.	Highly user-friendly, minimal effort required.

Table 2 shows a structured comparison among authentication system designs, based on five main dimensions: Accessibility, cost-effectiveness, energy consumption, usability, and security. Each constraint is assigned a value from 1 (Poor) to 5 (Excellent) as a comparative trade-offs in various biometric authentication modalities. This grid aids in the recognition of strengths and weaknesses in an attempt to make a balanced decision about security versus usable. It acts as an instructive tool for choosing the most suitable authentication scheme according to the requirements of the system.

$$weighted\ rate\ =\ (rate)(weighted\ percentage)$$

$$total\ \ rate\ =\ \sum weighted\ rate$$

Eq. 1. Weighted Average Formula

Furthermore, this equation is also employed to test and normalize the result of each authentication scheme on a 1-5 range. It computes the Rate as the ratio between the absolute (actual) performance of a design, and the maximum/minimum values it assumes by checking all the designs. The formula is designed to give the best design a score close to 5 and the worst design a score close to 1. This enables an impartial comparison of various authentication techniques according to different evaluation criteria.

Table 3. Multiple Design Constraints, Ratings, and Total Ratings

Design	Constraints	Rate	Weighted Rate	Total Rate
1	Global (25%)	3	0.75	4
	Economical (25%)	4	2	
	Environmental (25%)	3	0.75	
	Societal (25%)	2	0.5	
2	Global (25%)	2	0.5	5.75
	Economical (25%)	4	2	
	Environmental (25%)	3	0.75	
	Societal (25%)	5	2.5	
3	Global (25%)	4	2	7.25
	Economical (25%)	3	0.75	
	Environmental (25%)	4	2	
	Societal (25%)	5	2.5	

As shown in Table 3, the comparison of the three design alternatives points out their strengths and weaknesses uniquely depending on four main constraints: Global, Economical, Environmental, and Societal. All the constraints were equally weighted (25%) to make a balanced and unbiased comparison of the alternatives. The ratings for each design were based on how they performed under these categories, which represented their overall viability and effectiveness.

Design 1 scored a sum value of 4, implying a marginal performance. Economic constraint (4) received the highest score, and achieved the highest weight for its final score (2). This value shows that optimal option is design 2 in terms of saving resources and cost. This fact makes it an attractive choice for applications where costs are the governing consideration. But its Global (3), Environmental (3) and Societal (2) scores reveal its weaknesses. The design rating of 3 shows that the design has global potential, but it may require some adaptability for broader use. The Global rating of 3 indicates sustainability to some extent but with no significant breakthroughs in environmental impact reduction. The lowest score on all categories is the Societal score of 2 which shows that this design does not contribute in a useful manner to the well-being effect, accessibility or inclusivity and this can be a big disadvantage for projects that targeting on society.

Design 2 achieved a total score of 5.75, being significantly better than Design 1. What sets this one apart is being rated 5 in Societal, which added weight to the weighted aggregate (2.5). The rating indicates that Design 2 highly prioritizes societal impact, access, and inclusiveness. This could be the outcome of characteristics that benefit less-privileged communities, improved friendliness of use, or ethical considerations. While Design 2 received a lower Global of 2 as compared to 1, this implies it is not globally applicable, but applicable in certain areas or applications. Its Economical (4) and Environmental (3) scores are also the unchanged as Design 1, which would imply that while cost-effectiveness and sustainability standards is at air with Design 1, it does not show significant improvement there. The trade-off here, and this is achieved at the best in social gains, is at the expense of global flexibility.

Out of the three, Design 3 was the top choice with the highest overall score of 7.25. This design had high ratings in all four constraints, and it guaranteed a balanced and complete performance. Its Global rating of 4 means that this design has strong international scalability, indicating it is flexible to apply to various markets and uses. The Environmental score of 4 indicates a greater emphasis on sustainability, implying that the design has environmentally friendly materials, energy-efficient devices, or lower carbon footprint practices. The Societal score of 5 was the highest across all designs, reflecting its outstanding capacity to advance social welfare, inclusivity, and user engagement. The only category where Design 3 lagged behind was in the Economical category (3), suggesting that it could be more costly to implement than the other two designs. But the better scores in the other constraints counter this weakness, so Design 3 is the best-rounded and futuristic option.

Each has strengths and weaknesses, but it comes down to project priorities. If cost is the most important factor, then Design 1 would be the optimal solution as it results in the highest efficiency although contributors at the societal level are limited. If social benefits are the priority, Design 2 is the best choice as it gives a high priority to social benefits, although it is not well generalizable in the world. And if a balanced and comprehensive alternative is required, Design 3 is the best option, given that the decision must align with the total objectives of the project, whether they be weighted more in favour of the financial, social, or rated criteria. From this analysis, it is clear that Design 3 is better than the other two for providing the minimal value in each of the constraints. With its powerful global coverage, and sustainable and social relevance, it is the most efficient and comprehensive powertrain, guaranteeing long-term existence and general feasibility. Although more expensive in cost, its advantages make it the proposed architecture for implementing.

Trade-Offs

In designing and implementing a multimodal biometric authentication system that combines keystroke dynamics and speech biometrics, several trade-offs must be considered to optimize security, performance, cost, and user experience.

- **Security vs. User Experience**

  A more secure authentication method may be problematic for consumers who require numerous biometric inputs or longer processing period; less inconvenient techniques may be more accessible but less secure.

- **Accuracy vs. System Performance**

  A complex biometric system makes the authentication more accurate and reliable but it can affect the process and it can slow down the system. Also, it is better to use more computer resources compared to faster and simpler models.

- **Multi-Modal Fusion Complexity**

  To make the biometric more reliable and accurate it needs advanced methods but the system requires more processing power rather than some easier methods.

- **Adaptability vs. Security**

  An adaptive biometric system that adjusts to changes such as voice tone or typing speed is easier for users, but it may also be easier for attackers to intrude or hack the system.

- **Cloud-Based Processing vs. Privacy**

Cloud processing of biometric data offers stronger security choices but invades user privacy and needs a reliable internet connection, while local processing is safer from a privacy perspective, but less comprehensive.

Trade-off Considerations:

- **Strengthening Security Measures**

  Pros: It keeps unwanted users out and it also protects against anyone being able to impersonate or deceive the system.

  Cons: It would be a hassle to customers as it would require an additional step or a longer waiting time than normal just to log in.

- **Simplifying Authentication Process**

  Advantages: It simplifies and speeds up logging in, that may be the users much prefer to use.

  Cons: Making the system too simple makes it easily prone to attacks or intrusion.

- **Using a Complex Machine Learning Model**

  Advantages: It can make the system very reliable and accurate and it voids mistakes in recognizing people in authentication.

  Cons: More time to process and more powerful computers, which takes more time or costs more.

- **Using a Simpler Machine Learning Model**

  Pros: It works faster and doesn't need powerful machines, it can run on basic devices smoothly.

  Cons: It might lead to errors like letting the wrong person in or locking out the right one.

- **Using High-Sensitivity Matching**

  Pros: It can be good for security because it can catch more of the people who are trying to get in to the system.

  Cons: Possibly it might also block some legitimate users by mistake, which can be really frustrating.

- **Implementing High-Sensitivity Matching**

  Pros: It allows more real users to get through without being wrongly rejected, which makes the experience smoother and less frustrating.

  Cons: However, it can also let unauthorized users slip through more easily, which weakens the overall security of the system.

- **Implementing Low-Sensitivity Matching**

  Pros: It lets more real users and avoids unnecessary rejections and it's making the system more user-friendly.

  Cons: This can make the system less secure because it may allow people who shouldn’t have access to get in.

- **Cloud-Based Authentication**

  Pros: It can handle advanced features and manage large amounts of data easily.

  Cons: But, it raises concerns about privacy and it also needs a reliable internet connection to work properly.

- **On-Device Processing**

  Pros: Everything stays on the device so that it protects user data better because it works even without the internet.

Cons: It can be slower or limited because the device itself may not be powerful enough to do for complex tasks.

Trade-Off Result:

Multimodal fingerprint identification systems exchange security, accuracy, utility, and speed as a secure system might prevent illegal access but may be difficult to use. While simpler login techniques are quicker and easier, they may be less secure, but more likely to accept phony users or accidentally reject actual ones. Complex fusion and fingerprint algorithms improve accuracy and reduce errors, but they need more computer resources and slow down processes.

Given that they can handle small changes in biometric input, like type speed or face angle, adaptive biometric systems make the user experience better which also can make the system more open to fake attacks. Tougher security settings might better stop these kinds of hacks, but they could also keep real people from accessing the site. Cloud-based identification can handle a lot of data and complicated jobs, but it needs to be connected to the internet and makes people worried about data safety. Processing data directly on a device, on the other hand, protects user privacy and lets them access data when they're not online, but the hardware of the device may limit what they can do. Finding the right mix between these factors is the best way to go about things as it is important for a well-designed multimodal fingerprint system to be safe, reliable, easy to use, and adaptable to the needs of its application.

Engineering Standards

Biometric systems must conform to well-established security standards, making sensitive biometric data safe, ensuring accuracy of the system, and keeping authentication secure and valid

on different devices. The ISO/IEC 24745 standard defines a basic model for protecting biometric information and emphasizes methods of secure storage, access, and handling of biometric templates. In implementing this standard, safeguards for biometric data against unauthorized access will become paramount for systems depending on remote authentication or cloud storage.

While ISO/IEC 30107-3 has presented standards for Presentation Attack Detection (PAD) against a range of potential spoofing attacks on biometric systems, they offer applicability in contexts where counterfeit attacks involve the use of lifelike replicas and or digital manipulation to dupe the biometric system. Adhering to this standard will definitely increase the system robustness, thereby enhancing its usability in practical settings.

NIST Special Publication 800-63B deals with Digital Identity Guidelines and serves in an important way in the field of biometrics for authentication. This publication provides not only best practices for the deployment of biometrics within multi-factor authentication systems but also assurance levels for authentication. It gives support for organizations, especially sensitive sectors like banking, healthcare, and government, in aligning their security in accordance with the practices of the industry, thus complying with regulatory and security requirements.

ISO/IEC 19795 and NIST Biometric Performance Testing Standards further define the methodologies for being able to assess the performance of biometric systems. These standards define how one measures and calculates performance requirements like FAR and FRR so that biometric systems can achieve the very highest accuracy with minimum authentication error. By following these standards, the developers will enhance their algorithms to be most secure with the best user experience.

Also, ISO/IEC 27001 marks an important standard on Information Security Management System (ISMS) and gives procedures on how to protect biometric information from breaches, cyberattacks, or unauthorized alteration. Compliance with this standard ensures that the biometric system is set up with stringent policies of data security and that it shall provide strong protection against any potential threats.

ISO/IEC 19989-1:2020 concentrates on biometric presentation attack detection to constitute secure remote authentication and address security challenges encountered by online and mobile authentication. This standard assumes great importance owing to the growing application of biometric authentication for remote login into financial services, enterprise networks, and mobile applications, which can resist great spoofing advances. In addition, ISO/IEC 27035 prescribes the Information Security Incident Management guidelines-it describes the processes involved in detecting, managing, and recovering from security incidents. In terms of ensuring that the integrity and availability of biometric authentication systems are not compromised, especially in case of a breach, this is absolutely vital.

FIPS 140-2 defines Security Requirements for Cryptographic Modules that secure the transmissions of biometric data-stressing that any encryption used to secure data in transit and storage satisfy the highest security standards. This is very important when transmitting sensitive biometric data, especially when cloud-based systems are involved.

This means that the standardization provides enough guidelines for biometric authentication systems to necessarily be secure without compromising reliability, accuracy, or the experience of users. In addition to enhancing the security of systems, the application of these architectures would maintain the security of the precious biometrics in the process of providing

effective and safe authentication in different forms or types of platforms and devices. Such an invitation to both users and stakeholders builds trust, making the system impervious to security threats while providing smooth, effective, and real-world authentication experiences.

CONCEPTUAL FRAMEWORK

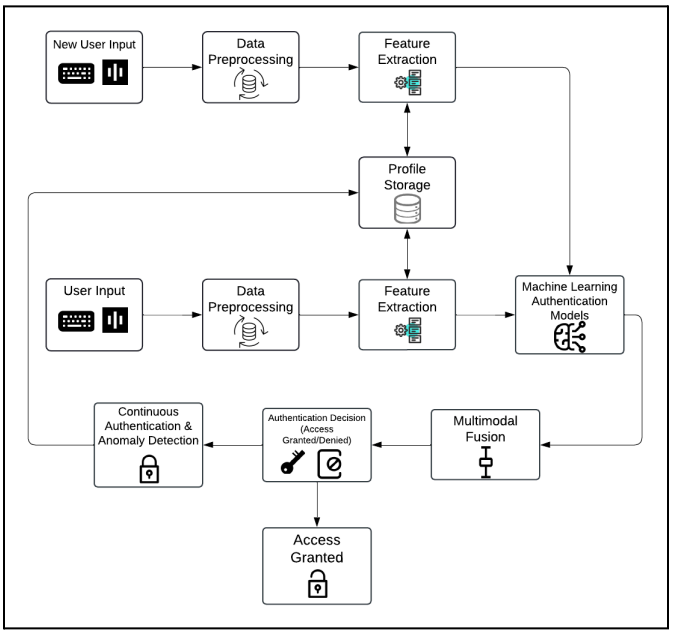

Figure 4. Diagram of the Proposed Authentication System

In figure 4, the diagram illustrates a Multimodal Biometric Authentication System that integrates keystroke dynamics and speech biometrics for secure authentication. It distinguishes between New User Registration and User Authentication (Login).

For New User Registration, the user provides keystroke and speech biometric data, which undergo Data Preprocessing and Feature Extraction to capture unique typing and voice patterns. These extracted features are securely stored in the Profile Storage database for future authentication. For User Authentication (Login), the user inputs new biometric data, which follows the same preprocessing and feature extraction steps. However, instead of being stored, it

is compared to the saved profile using Machine Learning Authentication Models like Random Forest (keystroke) and CNN (speech). The system then applies Multimodal Fusion, combining authentication scores from both biometric modalities to enhance accuracy.

The Authentication Decision Engine evaluates the results to either grant or deny access. If authentication succeeds, the user gains entry; otherwise, the Continuous Authentication & Anomaly Detection Module monitors ongoing user activity. If anomalies are detected, re-authentication is triggered. This structured process strengthens security by ensuring that even if one biometric factor is compromised, the system remains resilient against unauthorized access.

TECHNICAL DESIGN AND PROCEDURE

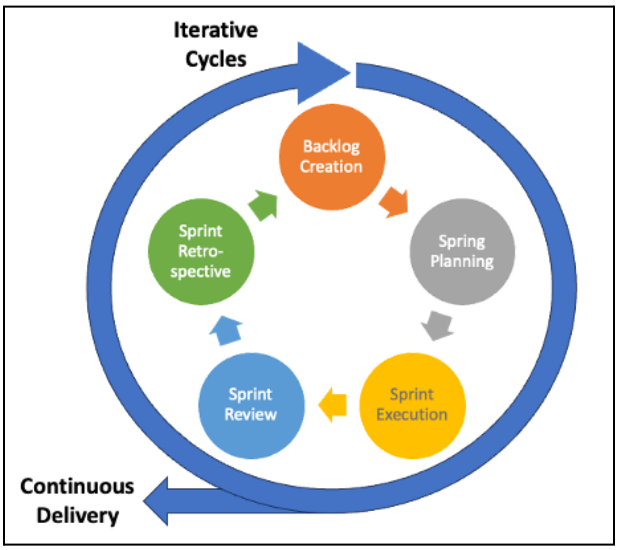

Figure 5. Agile Framework

As shown in figure 5, the development of the multimodal progressive authentication system follows the Agile methodology, structuring the process into five sprints, each designed to iteratively build, test, and refine key system features. For this project, Sprint 1 lays the

foundation by implementing mechanisms for keystroke and voice data collection, preprocessing, and feature extraction. Keystroke data is collected by timing how long each key is pressed and how often it is pressed. Voice data, on the other hand, is collected by focusing on pitch, rhythm, and tone. Both types of biological data are preprocessed before they are stored to get rid of noise and define entry forms which ensure that the data is accurate and consistent. Then, the data is kept safely using encryption methods to keep private fingerprint data from getting into the wrong hands.

Sprint 2 centers on keystroke-based authentication by developing an algorithm that compares a user's real-time typing patterns with their previously enrolled keystroke data. If a match is detected, access is granted; if not, the system transitions to Sprint 3, where voice recognition is introduced as a secondary method which involves analyzing the user's speech characteristics— such as pitch, tone, and cadence—to verify identity. It establishes a layered authentication flow where voice recognition acts as a backup when keystroke verification fails. Sprint 4 further strengthens the system by incorporating a fallback mechanism through security questions which provides users with a final authentication option in cases where both biometric methods fail. Also, an attacker-flagging feature is introduced, which monitors authentication attempts like if a user exceeds a specified number of failed attempts, they are flagged as a potential threat and access is restricted to safeguard the system from brute-force or impersonation attacks.

Lastly, Sprint 5 is all about integrating the whole system so that the voice recognition, security question, and typing dynamics units all work together without any problems which the users are asked to agree with the changes, and speed is improved. Real users' feedback is looked at to make the site easier to use and safer. By using this organized sprint method, the

development team makes sure that the system changes in small steps, responding to user needs and improving the security and dependability of authentication over time.

Development Process of the Multimodal Biometric Authentication System

The development of our Multimodal Biometric Authentication System using Keystroke Dynamics and Speech Biometrics will follow a structured process to ensure accuracy, security, and efficiency.

1. **Data Collection**

The first phase of the system is for data gathering, where users are asked to provide inputs and register their data on the basis of pre-defined fixed-text for keystroke dynamics and speech biometrics. For the keystroke data, data such as the time of release and press of keys, dwell time, and flight time are recorded when users type the pre-defined passphrase or words. This is done so that the same data is consistent across sessions. The users are then requested to speak a specific passphrase into a special microphone, so good-quality, consistent audio samples can be recorded. From the speech records, significant vocal features and MFCCs are determined to identify each speaker's unique vocal signature. With the standard and pre-provided input and spoken data texts, the system offers a contained framework for building a better and more consistent multimodal biometric model.

2. **Selecting the Best Algorithms for the Proposed Authentication System**

Choosing the right algorithms for the multimodal biometric authentication system using keystroke dynamics and speech biometrics is crucial to ensuring its accuracy, security, and

efficiency. To achieve this, different models will be tested and compared based on how well they meet the system's goals, particularly in terms of feature extraction, classification, and fusion.

Before deciding on specific algorithms, the system's key objectives must be clearly defined. These include achieving high authentication accuracy, keeping processing time low for real-time use, ensuring protection against spoofing attacks, and making the system adaptable for different users and devices. The models will be evaluated based on biometric authentication metrics like false acceptance rate (FAR), false rejection rate (FRR), equal error rate (EER), and processing time to determine which algorithms perform best.

For keystroke dynamics, several machine learning models will be considered. Support vector machines (SVMs) are great for small datasets and tend to provide high accuracy. Random forest (RF) is useful for handling noisy data and identifying complex typing patterns. K-nearest neighbors (KNN) is easy to implement but can be slow when working with large datasets. Meanwhile, recurrent neural networks (RNNs) and long short-term memory (LSTM) networks are effective for analyzing typing behavior over time, although they require more processing power. The performance of these models will be tested using cross-validation, and the one with the lowest EER and fastest processing time will be selected.

For speech biometrics, different algorithms will be analyzed to determine the best approach for verifying users based on their voices. The MFCC + HMMs is a traditional method that works efficiently for speech recognition. Convolutional neural networks (CNNs) can capture complex voice features, while recurrent neural networks (RNNs) and LSTMs are well-suited for analyzing voice sequences, although they require more computational resources. These models

will be tested using recorded voice data to find the one that offers the best combination of performance and efficiency.

To combine both biometric methods, different fusion techniques will be explored. Rule-based fusion assigns specific weights to each biometric input, while machine learning-based fusion, like decision trees or logistic regression, optimizes authentication by learning from the data. Deep learning-based fusion, such as using neural networks or transformer models, can provide higher accuracy but requires a large dataset for training. Several fusion strategies will be tested to find the most effective approach that minimizes EER while ensuring fast and reliable authentication.

Finally, the system will undergo thorough testing with real users under different conditions, such as background noise affecting voice inputs and variations in typing speed. Stress tests will also be conducted to evaluate its resistance to spoofing attacks. The best algorithm combination will be chosen based on its ability to deliver high security, low error rates, and real-time performance.

3. System Integration

Following model training, the system is integrated and deployed as an end-user-friendly authentication system. Users are prompted to input a passphrase through typing and voice input, which is compared and analyzed against biometric templates stored. Authentication success is determined by a predefined confidence level such that both biometrics have to match before access can be obtained. The system is simplified to facilitate user interaction with the authentication system.

4. Testing and Optimization

Performance testing is conducted for various reasons to ensure unbiased performance and reliability of the Multimodal Biometric Authentication System. The system is assessed on the following important performance measures: false acceptance rate (FAR), false rejection rate (FRR), and equal error rate (EER). These performance measures play a role in reaching a balance between security and usability by finding how often unauthorized access is being erroneously provided or legitimate users are being denied. Variations of these values are monitored over time to adjust authentication models for their best efficiency.

Testing also receives feedback from the users in the system development stage to refine system responsiveness and overall usability. Authentication speed i.e. the speed versus accuracy tradeoff is analyzed to limit delays. The feature extraction process has been optimized to better allow the system to distinguish between users. Keystroke and speech recognition machine learning models are fine-tuned through iterative cycles of parameter optimization and the provision of additional training datasets to ensure maximum classification accuracy.

Reliability is enhanced by the improving fusion decision mechanism that integrates keystroke dynamics and speech biometrics. Different fusion mechanisms, from weighted score aggregations to machine-learning-based combination methods, are put to the test for determining the best fusion scheme. The system was also put to stress testing in terms of environmental conditions, such as varied background noises and typing conditions, to evaluate its robustness. With all these improvements, the system is further refined to have an effective, secure, and user-friendly authentication experience in different real-life scenarios.

5. Security and Deployment

In order to offer both security and accessibility, the system requires a flexible storage facility that can effectively accommodate authentication. Local storage enables fast and secure authentication offline, with users having fast access to accounts. Local storage, though, limits flexibility since it will force users to re-register when using devices. Cloud storage, in turn, provides access from various devices and backup of data but demands stringent security measures in order to avoid any unauthorized access. To override these strengths and weaknesses, a hybrid system for storage can be utilized. It refers to storing particular parts of the authentication locally in order to speed up verification and saving other information securely in a cloud database to provide seamless authentication.

In such hybrid deployments, network overhead will also be taken into consideration as transmission of biometric data may introduce delay and affect real-time performance. To avoid this, the system will use light-weight data formats, local pre-processing, and optimized transmission protocols to reduce delay and ensure responsive authentication in spite of network fluctuations

To prevent tampering with biometric data, integrity verification techniques are utilized to ensure that data stored is not tampered with but remains authentic. Other security features such as encrypted data streams, authentication logs, and anomaly detection functionality aid in tracking and preventing unauthorized access. By incorporating the above security measures, the system is secure but easy to use with the user able to authenticate from anywhere while their biometric data is safe. After all security functions are in place, the system then passes final testing for usability before fully deploying, making it functional in actual practice.

In addition, the following security testing methods will be employed to make sure that the system is strong and resistant to potential attacks:

1. **Penetration Testing.** This test will involve the simulation of cyber-attacks in an attempt to identify weaknesses in the biometric authentication system. This will cover the testing of front-end interfaces and back-end structures, like databases and APIs, to determine whether they are resistant to common attacks like SQL injections, brute-force attacks, and man-in-the-middle attacks.

2. **Vulnerability Testing.** This test is going to focus on the detection of known system security vulnerabilities. Automated vulnerability scanners will scan for weaknesses in configurations and codebase, while manual audits will address more complex security concerns to ensure the system is secure against latest threats.l cover more intricate security issues to make sure the system is safe against current threats.

3. **Authentication Testing.** This test will evaluate the performance and correctness of the authentication procedure. Given measures, such as False Acceptance Rate (FAR), False Rejection Rate (FRR), and Equal Error Rate (EER), will be used to check how effectively the system can authenticate genuine users and reject unauthorized attempts of access.

With these comprehensive security test processes, the system will be tested to ensure that it is secure, stable, and user-friendly in every environment.

Software Design

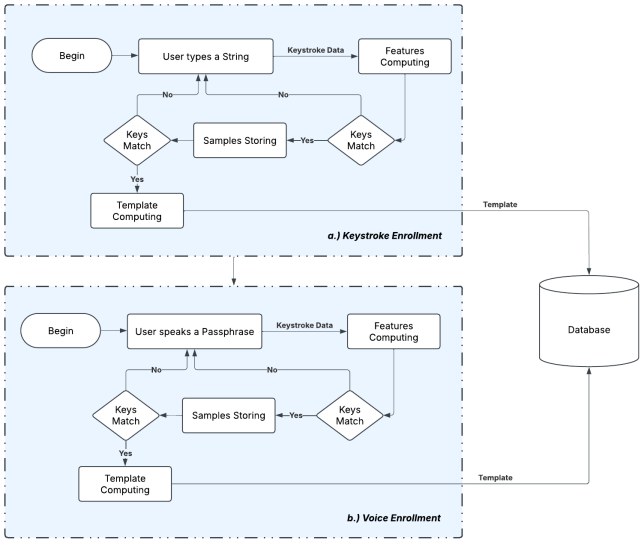

Figure 6. Flowchart of Enrollment Process

Figure 6 shows the enrollment process of the proposed multimodal biometric authentication system that combines keystroke dynamics and speech biometrics. During the (a)keystroke enrollment, a user is requested to type a predefined text, and the system acquires keystroke data, calculating relevant features like typing speed and key press duration. If the sampled samples are within specs, a keystroke template is computed and stored in the database, otherwise sample collection continues. Similarly, during (b) voice enrollment, the user utters out a specified passphrase, and speech features like pitch and frequency are extracted by the system. The process continues as in keystroke registration through verification and template computation prior to storage of the voice template in the database. Both biometric templates are used for future authentication, ensuring secure and efficient multimodal verification.

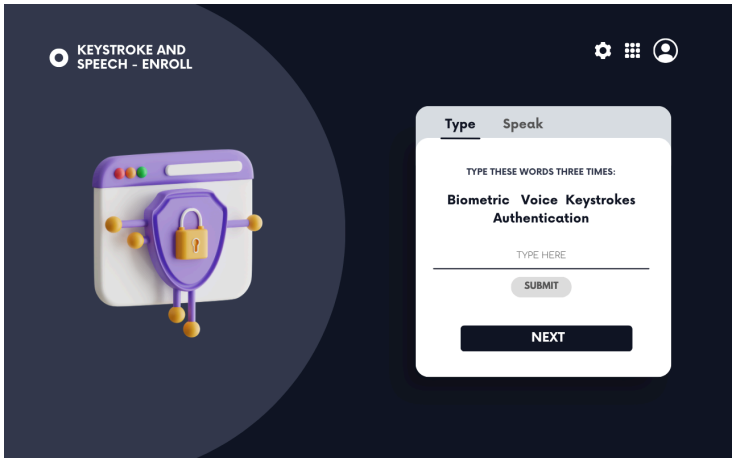

Figure 7. Biometric Keystrokes Enrollment

Figure 7 showcases the Keystroke Authentication Enrollment Interface, allowing users to input a predefined text multiple times to capture their unique typing patterns. The system records keystroke dynamics such as dwell time, flight time, and typing speed to generate a biometric profile.

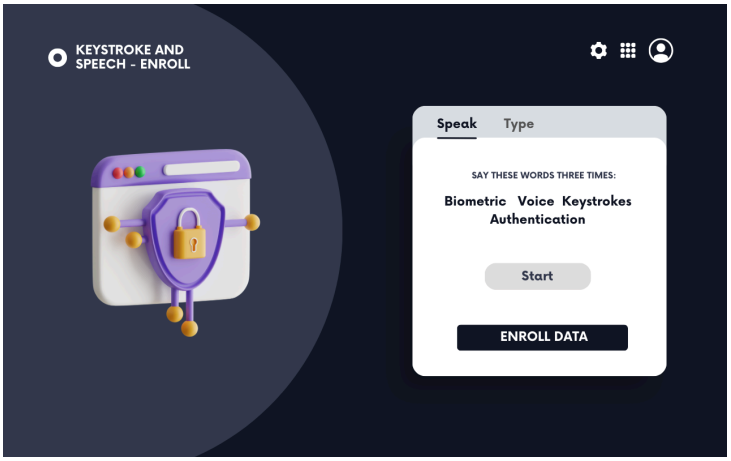

Figure 8. Biometric Voice Enrollment

Figure 8 presents the Voice Authentication Enrollment Interface, enabling users to enroll their speech biometrics by repeatedly speaking a predefined passphrase. This step ensures that the system captures unique vocal characteristics, including pitch, tone, and rhythm, which will be used for future authentication.

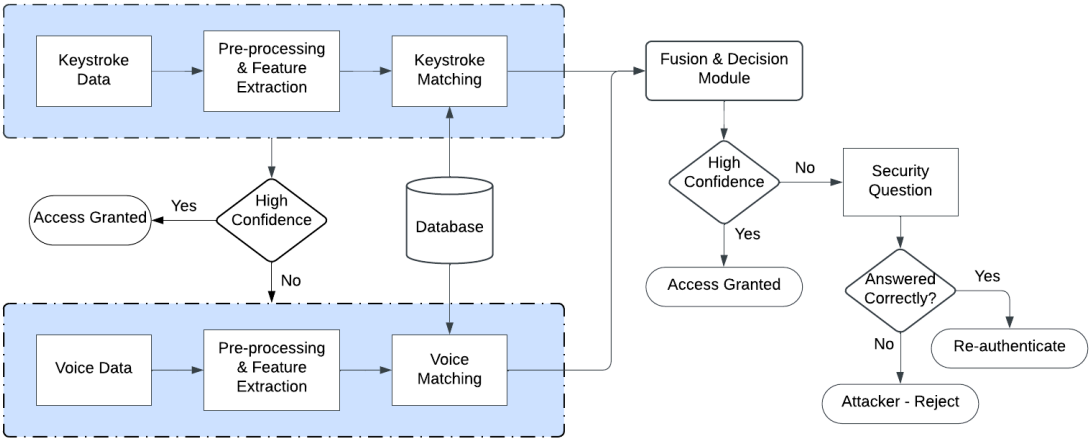

Figure 9. Flowchart of Verification Phase

As shown in figure 9, the verification phase flowchart illustrates the authentication process using a multimodal biometric system that combines keystroke dynamics and voice recognition. The system first collects keystroke and voice data, followed by pre-processing and feature extraction. Each modality undergoes separate matching against stored templates in the database. If either the keystroke or voice modality achieves high confidence, access is granted. If not, both results are fused in a decision module for final evaluation. If the fusion still lacks confidence, a security question is presented as a fallback authentication step. If answered correctly, the user is prompted to re-authenticate, but if incorrect, access is denied, preventing attackers from bypassing the system. This layered approach enhances security by integrating behavioral and physiological biometrics.

EQUIPMENT/FACILITIES/PROGRAMS

This project does not require the use of major specialized facilities or equipment. However, several essential programs, software tools, and computational resources will be utilized to ensure the successful completion of the research. The project "Multimodal Biometric Authentication Using Keystroke Dynamics and Speech Biometrics as an Alternative to Traditional 2FA" will rely primarily on freely available software tools and resources. Keystroke dynamics will be analyzed using Python libraries like pynput and keyboard, while speech biometrics will be processed with tools such as PyAudio and SpeechRecognition. Machine learning frameworks, such as TensorFlow, Keras, and scikit-learn, will be used for developing and training the authentication models. For data visualization, tools like Matplotlib and Seaborn will be employed. Additionally, Git and GitHub will be utilized for version control, while Slack and Google Drive will support communication and file sharing among the project team. All of these programs and resources are freely accessible or open-source, ensuring that no additional financial investment is needed for specialized software or equipment.

DATA COLLECTION AND TREATMENT

a. Data Collection

The data collection will focus on collecting data for both keystroke and speech biometrics. It involves gathering the data through a controlled input process. To ensure the fairness and diverse dataset, the participants will be selected from different demographics depending on their typing speed, accents, and speech patterns. They are expected to perform tasks under different circumstances to ensure a seamless data collection.

The collection of keystroke data will involve the participants to input their data into the system where it extracts features including flight time, dwell, time, and typing speed. Their data will be recorded using a keystroke logging tool where they are tasked to repeatedly type predefined text samples. Meanwhile, speech data collection will include the voice pitch, tone, and pronunciation to be captured from participants by speaking in a high-quality microphone. The captured speech biometric data will be processed into Mel-Frequency Cepstral Coefficient (MFCC) for extraction of features.

After the collection of raw data, it will be stored in the database for further analysis. The dataset will be split into training, validation and testing subsets to train for evaluation and evaluate the authentication model effectively.

b. Data Treatment

A series of data quality-checking and consistency-checking algorithms will be used on the collected data before it can be integrated into the overall system. For keystroke dynamics, outlier detection may be performed on the raw data using methods such as the interquartile range (IQR) to get rid of extreme values that could affect the performance of the model. Normalization techniques will also then be used to ensure that typing speed and keypress duration are normalized, such that variations between users are adequately scaled. The performance of this preprocessing step will further enhance the comparability of keystroke patterns across users.

For speech biometrics, the collected audio data will be denoised and enhanced via spectral subtraction and voice activity detection (VAD) to get rid of unwanted noise and irrelevant audio segments. The cleaned audio signals will then be transformed to Mel-frequency cepstral coefficients (MFCCs), a popular feature extraction technique for speech processing. To make the model even more robust, augmentation techniques will be employed such as pitch

shifting and time stretching to emulate phrases spoken under different environmental conditions or natural variations in voice.

In order to keep its integrity and security, all processed data will first be encrypted before storage in the database of the system. The anonymization processes will also be put in place to remove any personal identifiers from the dataset to comply with data privacy laws. The cleaned and transformed dataset will then be divided into three subsets: Training (70%), Validation (15%), and Testing (15%). This structure will facilitate the learning of the model while limiting overfitting. This is a rigorous approach to data treatment that guarantees high accuracy and reliability in authenticating users by the system.

BUDGET REQUIREMENTS

This study is concerned with software-based methods that do not involve any hardware or physical infrastructures. With this, the expense of this study is relatively low. Special biometric scanners, lab equipment on a permanent installation, and other facilities are not necessary because this study is computational, something that would otherwise increase its cost. The biometric authentication system will be developed and tested using pertinent available data, templates, and software simulations rather than hardware, thereby significantly lowering costs.

EXPECTED OUTPUT

Multimodal authentication system output is designed to give users a secure and seamless experience with the help of both speech and keystroke authentication. The system aims to

identify the users based on their individual typing patterns as well as voice patterns, providing security and usability.

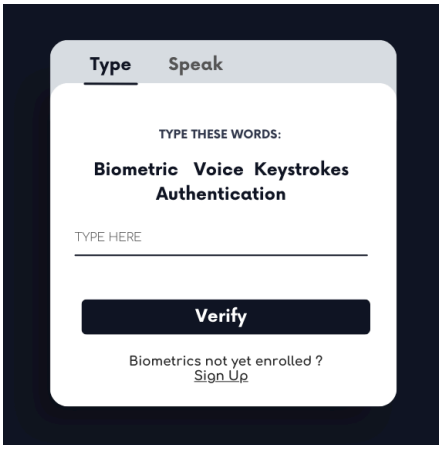

Figure 10. Biometric Keystrokes Authentication

Figure 10 presents the Biometric Keystroke Authentication interface, which is a key component of the multimodal authentication system. This section allows users to verify their identity using keystroke dynamics and speech biometrics.

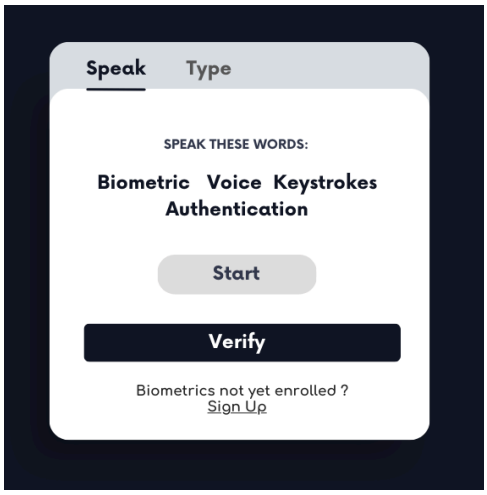

Figure 11. Biometric Voice Authentication

Figure 11 illustrates the Biometric Voice Authentication interface, a key component of the multimodal authentication system. This section of the system allows users to authenticate their identities through voice biometrics, examining their individual voice characteristics. "In Speak" , users are asked to recite a particular phrase, which is recorded and examined to compare with the enrolled biometric signature.

The system comes with a "Verify" button to facilitate processing of authentication and a "Sign Up" feature for users who have not yet registered in the biometric system. This image shows how user-friendly the design is and how easy it is to access the biometric authentication system while keeping security intact.

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